

CHAPTER-5: Implementation

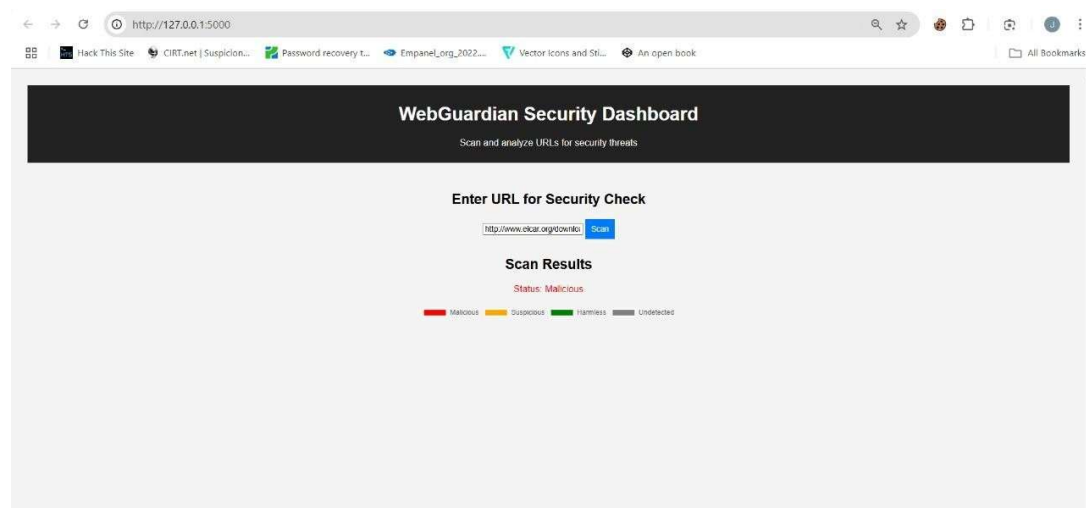
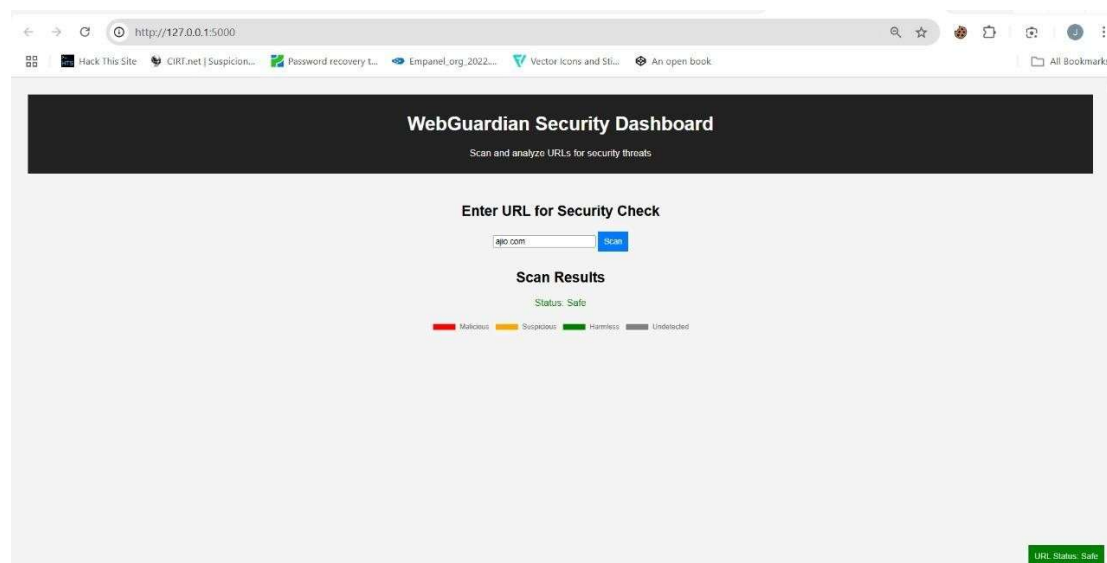
5.1 Overview

This module helps detect malicious URLs and domains. Currently, we use VirusTotal API to check URLs, and in the future, we will integrate Machine Learning for improved detection.

5.2 Current Implementation (Using VirusTotal API)

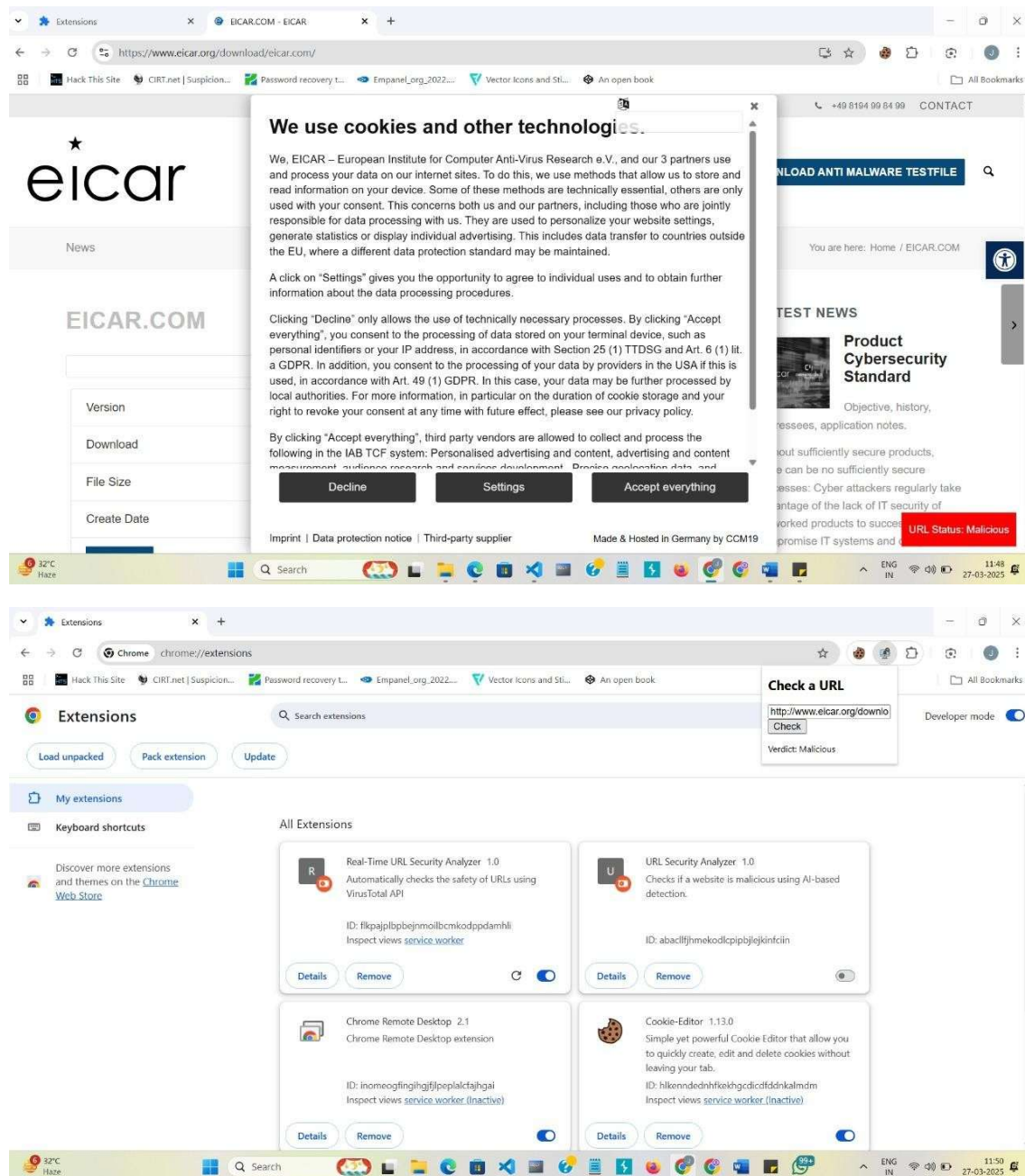
5.2.1 How It Works

1. User enters a URL or domain.
2. The system sends it to VirusTotal API.
3. VirusTotal returns a security report.
4. Results are displayed to the user.



5.2.2 Steps We Implemented

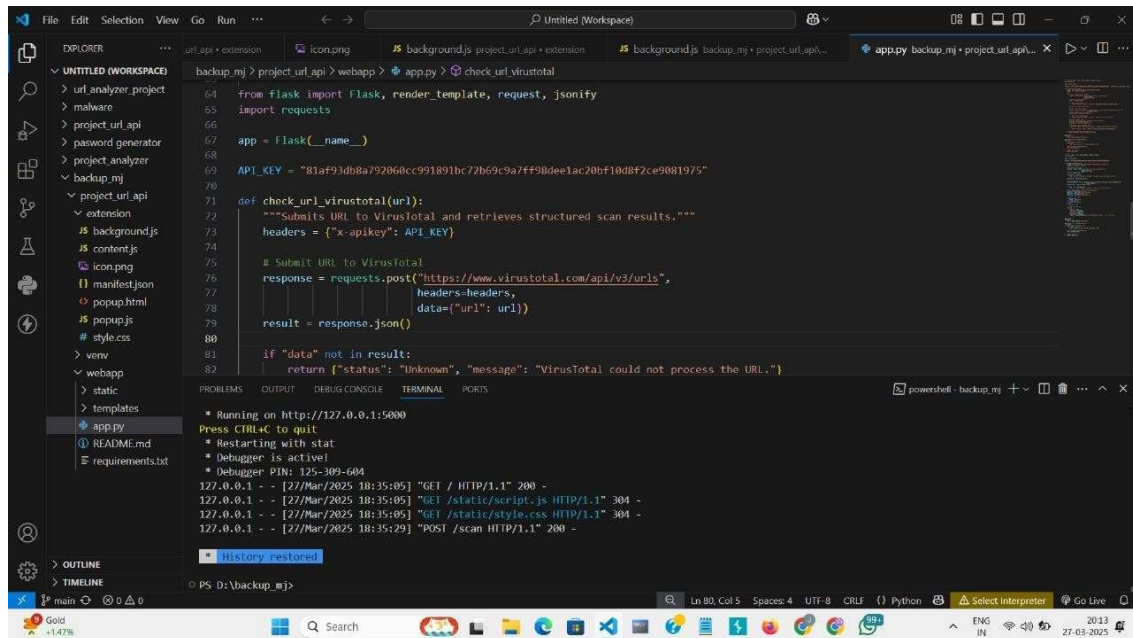
- ✓ Extract domain from URL
- ✓ Send request to VirusTotal API
- ✓ Retrieve scan results
- ✓ Display results in frontend



5.2.3 Code Implementation

1. User enters URL/domain in the frontend.
2. Backend (Flask) sends request to VirusTotal API.

3. Response is parsed, and the result is displayed.

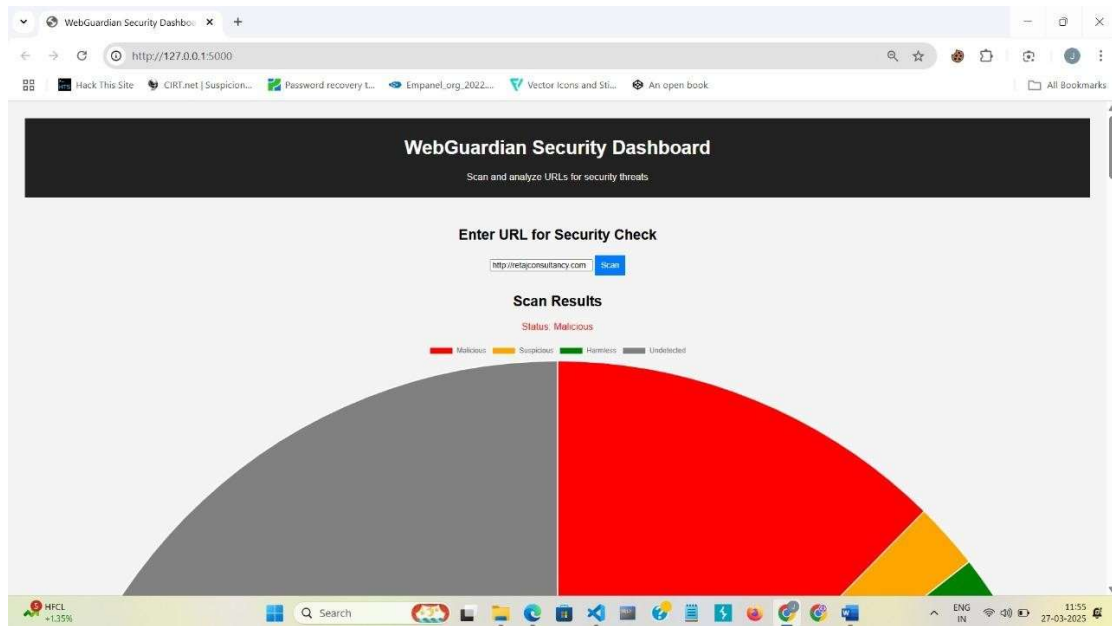


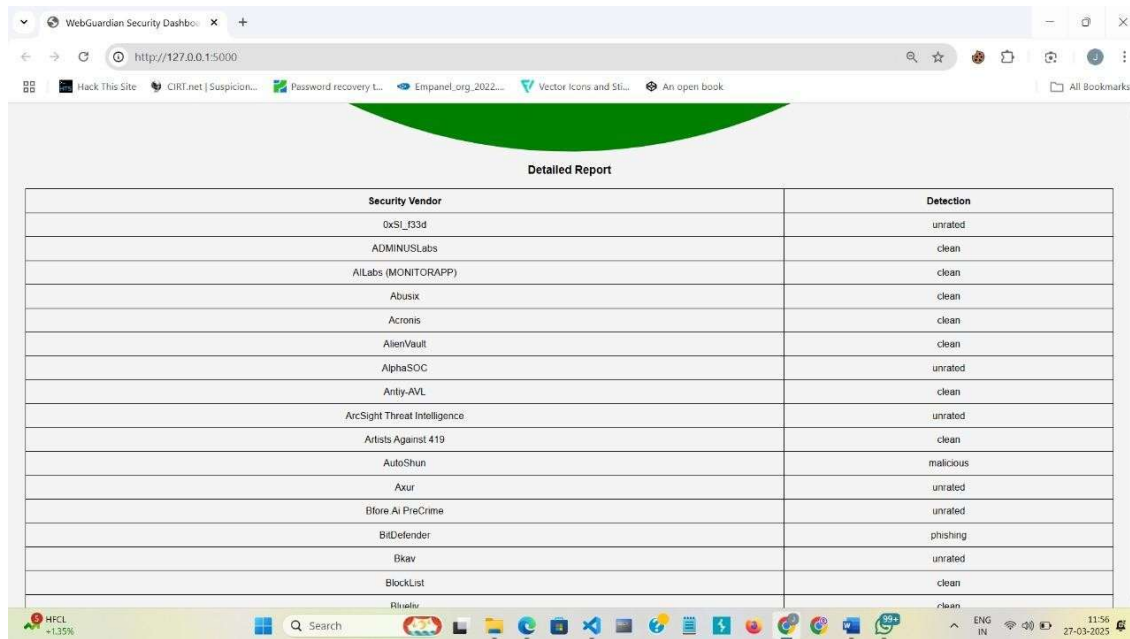
The screenshot shows a VS Code editor with a Python Flask application. The file explorer on the left shows a project structure with files like `background.js`, `content.js`, `icon.png`, `manifest.json`, `popup.html`, `popup.js`, `style.css`, `webapp`, `static`, `templates`, `app.py`, `README.md`, and `requirements.txt`. The main editor shows the `app.py` file with the following code:

```
64 from flask import Flask, render_template, request, jsonify
65 import requests
66
67 app = Flask(__name__)
68
69 API_KEY = "81af93db8a792060cc991891bc72b69c9a7ff98dee1ac20bf10d8f2ce9081975"
70
71 def check_url_virustotal(url):
72     """Submits URL to VirusTotal and retrieves structured scan results."""
73     headers = {"x-apikey": API_KEY}
74
75     # Submit URL to VirusTotal
76     response = requests.post("https://www.virustotal.com/api/v3/urls",
77                             headers=headers,
78                             data={"url": url})
79     result = response.json()
80
81     if "data" not in result:
82         return {"status": "Unknown", "message": "VirusTotal could not process the URL."}
```

The terminal at the bottom shows the application running on `http://127.0.0.1:5000`. It displays messages for starting the server, restarting with stats, and the debugger being active. It also shows the debugger PIN: `125-309-604`. The terminal output includes several HTTP requests and responses:

```
127.0.0.1 - - [27/Mar/2025 18:35:05] "GET / HTTP/1.1" 200 -
127.0.0.1 - - [27/Mar/2025 18:35:05] "GET /static/script.js HTTP/1.1" 304 -
127.0.0.1 - - [27/Mar/2025 18:35:05] "GET /static/style.css HTTP/1.1" 304 -
127.0.0.1 - - [27/Mar/2025 18:35:29] "POST /scan HTTP/1.1" 200 -
```





Detailed Report

Security Vendor	Detection
0xSI_03d	unrated
ADMINUSLabs	clean
AILabs (MONITORAPP)	clean
Abusix	clean
Acronis	clean
AlienVault	clean
AlphaSOC	unrated
Anity-AVL	clean
ArcSight Threat Intelligence	unrated
Artists Against 419	clean
AutoShun	malicious
Axur	unrated
Bfore AI PreCrime	unrated
BitDefender	phishing
Bkav	unrated
BlockList	clean
Broadcom	clean

5.2.4 Conclusion

- ✓ **Current Implementation – VirusTotal API for scanning URLs.**
- ✓ **Future Enhancement – Machine Learning for detecting new threats.**